

THE RESPIRATORY SYSTEM

Known as the *pulmonary system*, (*spir/o* = to breathe) functions *breathing* or *ventilation*.

A **pulmonologist** (*pulmon/o* + *-logist*) is a specialist in the anatomy, physiology, and pathology of the lungs, while a **respiratory therapist** evaluates and treats respiratory disorders.

STRUCTURES OF THE RESPIRATORY SYSTEM

Diaphragm; separates the abdominal cavity from the *thoracic* (*thora/o* = chest) cavity.

Pleura: membrane that surrounds the lungs

Lungs; a pair of large, spongy organs used for gas exchange the right lung has three lobes while the left lung has two lobes.

Pharynx: throat

Larynx: voice box

Combining Form	Meaning	Word Association
<i>bronch/o,</i>	bronchi	A <i>bronchodilator</i> is a medication
<i>laryng/o</i>	larynx (voice box)	<i>Laryngospasm</i> is the contraction of the vocal folds.
<i>nas/o, rhin/o</i>	nose	The <i>nasopharynx</i> refers to the upper part of the throat behind the nose.
<i>pharyng/o</i>	pharynx (throat)	Pharyngitis is the inflammation of the pharynx, “sore throat.”

pneu/o, pneu- mon/o, lung

Pneumonia is the infection of one or both lungs.

trache/o trachea

Tracheostomy is a surgical procedure

Two of the most common noninvasive measurements used are.

1-Oximetry: measurement of the blood oxygen saturation in the arteries by a device of *oximeter*.

2-Spirometry (spiro + metry): measurement of airflow taken in and exhaled from the lungs by a device of *spirometer*.

The suffix *-pnea* denotes breathing.

List for medical terms pertaining to respirations.

<i>eupnea</i>	<i>eu- (normal)</i>	eu + pnea	normal respirations
<i>dyspnea</i>	<i>dys- (bad)</i>	dys + pnea	difficult breathing
<i>apnea</i>	<i>a- (absence)</i>	a + pnea	temporary absence of breathing
<i>orthopnea</i>	<i>orth/o- (straight)</i>	ortho + pnea	difficulty breathing in any position
<i>bradypnea</i>	<i>brady- (slow)</i>	brady + pnea	less than 12 breaths/min.
<i>tachypnea</i>	<i>tachy- (fast)</i>	tachy + pnea	respirations more than 20 breaths per minute

Terms are frequently used in relation to the respiratory system:

hyperventilation: excessive ventilation of the lungs

hypoventilation: insufficient exchange of oxygen and carbon dioxide in the lungs

hypoxia: diminished oxygenation to the tissues

anoxia: extreme form of hypoxia, severe lack of tissue oxygen

SURGICAL AND THERAPEUTIC INTERVENTION

Endotracheal intubation (endo = inside + trache/o + al): insertion of artificial airway into the trachea.

nasotracheal intubation (nas/o = nose): insertion of an airway tube through the nose.

orotracheal intubation (or/o = mouth): insertion of an airway tube through the mouth.

Ventilator an apparatus used to administer artificial respiration in cases of respiratory failure.

Most common surgical procedures related to the respiratory system.

Surgical Procedure	Word Parts	Definition
<i>tracheotomy</i>	trach/o + tomy	creation of an opening in the trachea
<i>tracheostomy</i>	trach/o + ostomy	creation of artificial airway in the trachea
<i>biopsy</i>	bi + o + o/psy	puncturing the skin to obtain a sample of the tissue
<i>percutaneous</i>	per + cutane/o	
<i>pneumonectomy</i>	pneumon/o + ectomy	removal of all or part of the lungs

Lobectomy

lob/o + ectomy

removal of a lobe of the lung

Rhinoplasty

rhin/o + plasty

surgical repair of the nose

Pneumocentesis

pneum/o + centesis

puncture of the lungs to remove fluid

thoracentesis

thorac/o + centesis

puncture of the chest cavity to drain fluid

THE DIGESTIVE SYSTEM

The *digestive system* is responsible for the provision of water, nutrients, and minerals to the body. **Alimentation** (to nourish) is the term used for the process of giving or receiving nutrition, while **metabolism** is used to describe all the body processes involved in maintaining life.

There are four processes involved in digestion:

Ingestion: taking foods or fluids orally.

Digestion: physical and chemical breakdown of food into smaller components that can be absorbed in the bloodstream.

Absorption: passage of food molecules from the small intestine into the blood and lymph capillaries.

Elimination: removal of food particles that were not digested through *defecation*.

Word parts related to digestion and nutrition:

Word Part	Meaning	Word Association
-ation	action or process	<i>Defecation</i> is the process of passing out stool or feces via the anus.
bil/i, chol/e	bile	The <i>biliary</i> system consists of the liver, gall-bladder, and the bile duct.
cirrh/o	orange-yellow	<i>Liver cirrhosis</i> refers to the chronic scarring and damage to the liver.
-orexia	appetite	<i>Anorexia</i> is the loss of appetite.
-pepsia	digestion	<i>Dyspepsia</i> is the indigestion or painful and upset stomach.

Alimentary Tract

The *alimentary tract* = *digestive tract*, starts from the mouth and continues down to the anus.

Accessory Organs of Digestion

Word Part	Meaning	Word Association
<i>cholecyst/o</i>	Gallbladder	<i>Cholecystectomy</i> is the surgical removal of the gallbladder.
<i>hepat/o</i>	Liver	<i>Hepatitis</i> refers to the inflammatory condition of the liver.
<i>pancreat/o</i>	Pancreas	<i>Pancreatography</i> is the radiographic examination of the pancreas.

Hyperglycemia may lead to the following signs and symptoms (*poly* = excessive):

Poly + phagia (eating)

Poly + uria (urination)

Poly + dipsia (thirst)

The most common type of feeding tubes is:

Nasogastric: naso + gastr/o + ic.